Tyler K. Akana

253-228-6953 | Tyler.Akana@outlook.com | LinkedIn

Experience

Engineering Analyst @ DB Engineering | May 2025 – Present

- Analyzed HVAC and mechanical equipment serving the Microsoft campus, identifying performance issues and quantifying maintenance-related cost savings.
- Performed commissioning of HVAC and mechanical systems for commercial clients, reviewing sequences of operation, mechanical and electrical drawings, and air balance diagrams; developed and executed functional performance tests (FPTs) to validate equipment and system operation in the field.
- Developed automated monitoring and reporting tools to analyze equipment performance and support engineering decisions.
- Worked as a team lead on a water-demand study across 95 buildings on the Microsoft campus, integrating equipment inventories, metering, and operational data to identify millions of gallons in projected savings.

PACCAR-Kenworth Design Project | Aug 2024 – Dec 2024 | **Portfolio**

Pneumatic Force Measurement System for Semi-Truck Hood Lift Testing.

- Designed and integrated a pneumatic test system for heavy-duty truck hood lift testing, using SOLIDWORKS simulation to validate a 600 lbf design.
- Developed an Arduino-based data acquisition system costing <10% of commercial alternatives, including instrumentation, test procedures, and technical documentation.
- Collaborated with PACCAR engineers to refine the design and present development results.

Undergraduate Research @ Washington State University | Jan 2024 – Feb 2025 | **Portfolio**

Heat Box Project: Thermal-Fluid System for Agricultural Heat Stress Analysis

- Designed, prototyped, and tested a mechanical–fluid system from concept through deployment.
- Used CFD and analytical methods to size components and achieve required airflow and pressure characteristics.
- Developed a wireless temperature-monitoring system for real-time data acquisition and remote performance evaluation.

Skills

SOLIDWORKS, FEA, CFD, Pneumatic Systems, Experimental Design, Instrumentation/DAQ, Arduino/C++. Matlab/Python, DFMA, PowerBI, SQL.

Education

Washington State University, B.S. Mechanical Engineering